

Reporte de Análisis Físico

Validación MC y Asimetría UMD/SD

Configuración de la Simulación

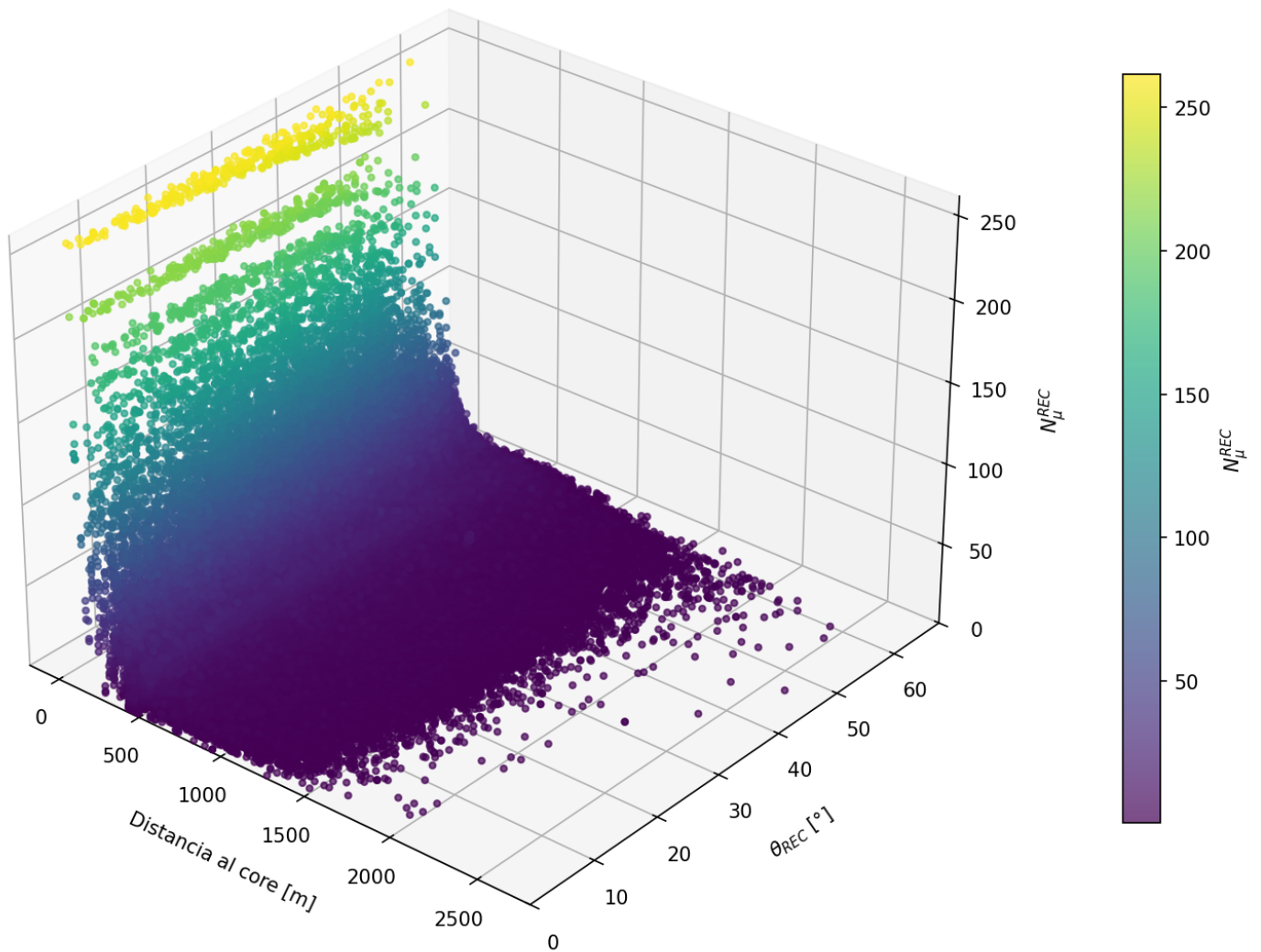
Parámetro	Valor
Modelo Hadrónico	SIBYLL 2.3d
Primario	Protón
Rango Energía	17.5 - 18.0 log(eV)
Carpeta	
Eventos Totales	4,969
Módulos Totales	1,589,487

1. Limpieza de Datos

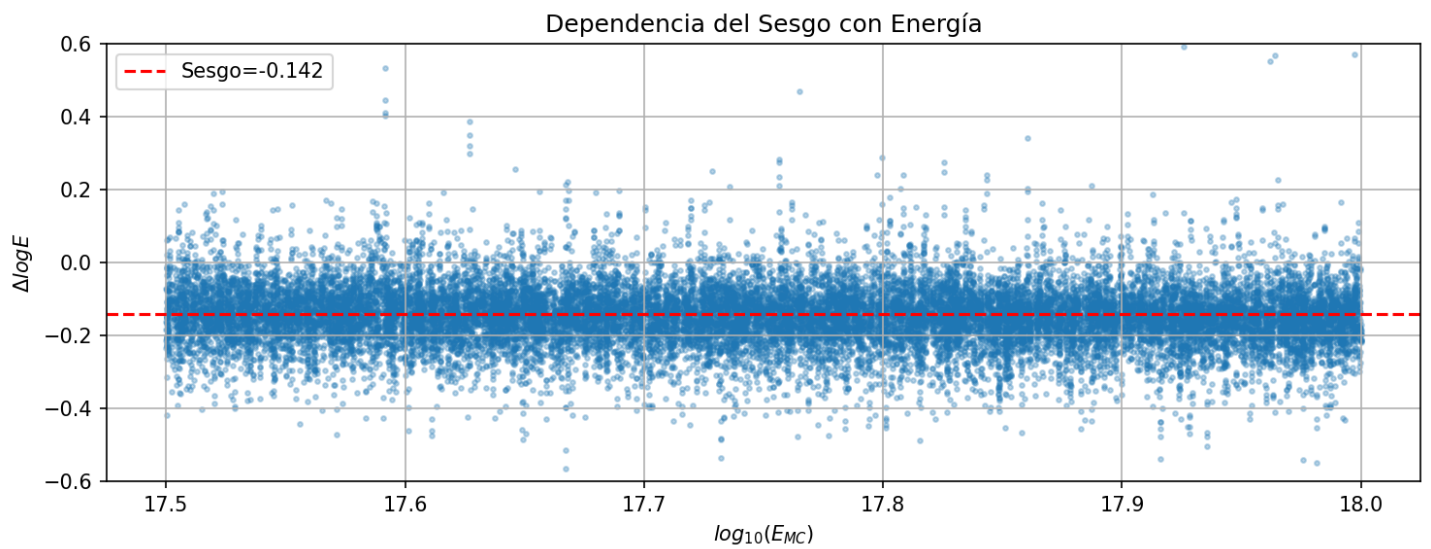
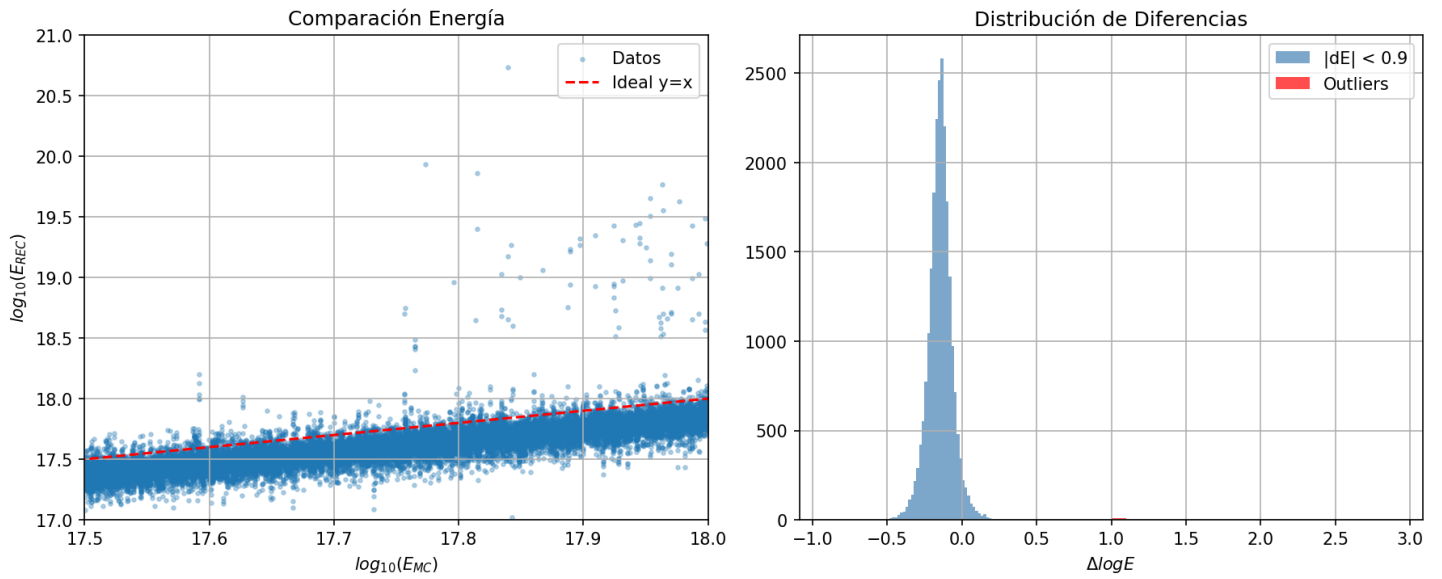
REPORTE DE LIMPIEZA

Métrica	Valor
Filas Iniciales	1589487
Filas Eliminadas	56658 (3.56%)
Filas Finales	1532829
Causa Principal	NaN en logE_REC (56658)

2. Visualización General

N_{μ}^{REC} vs distancia y θ 

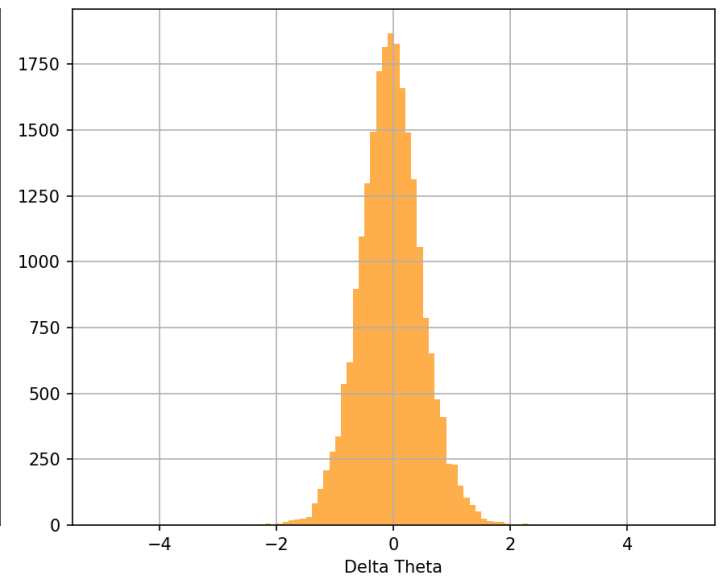
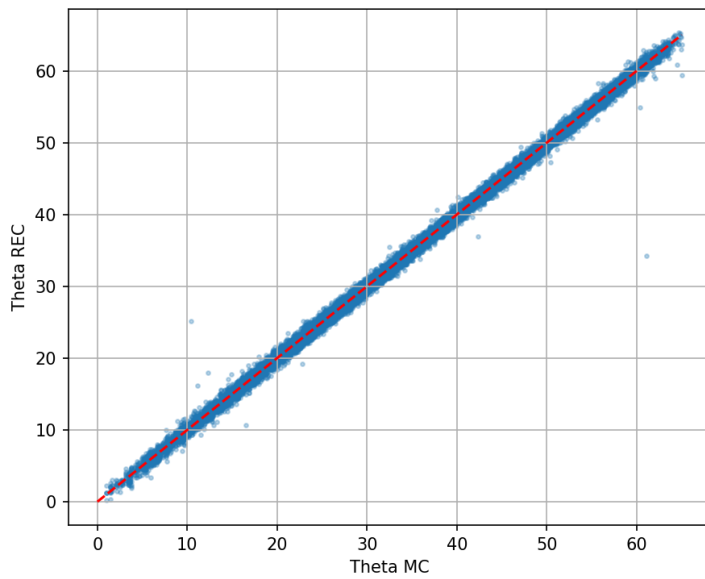
3. Resolución de Energía



Sesgo (Mean): -0.1420
 Resolución (Std): 0.1112
 Outliers (> 0.9): 47 (0.20%)

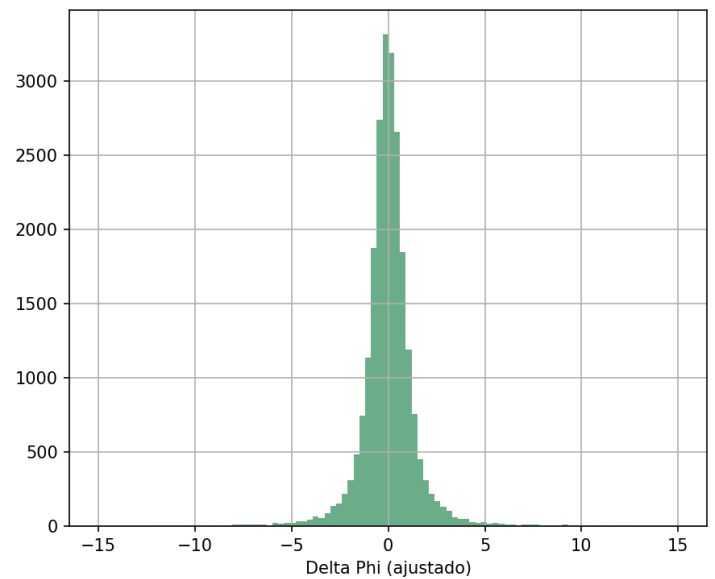
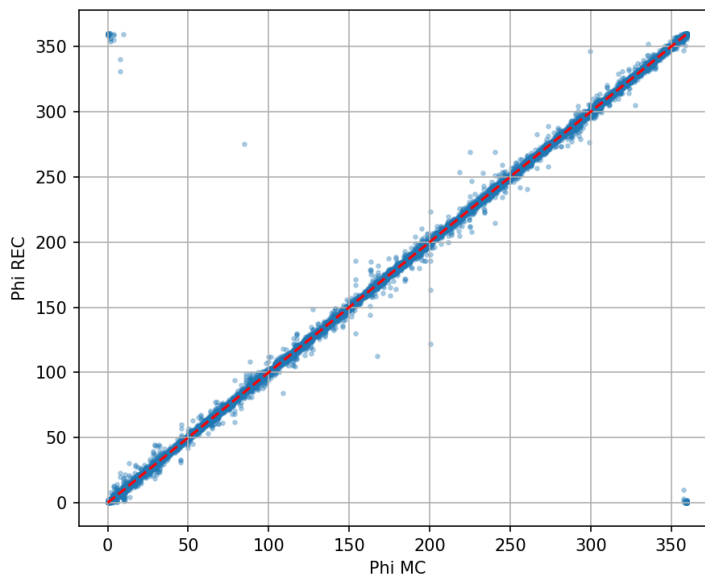
4. Resolución Angular

Theta (Cenital)



Sesgo: -0.0468 deg
Resolución: 0.5787 deg
Outliers ($> 10.0^\circ$): 2 (0.01%)

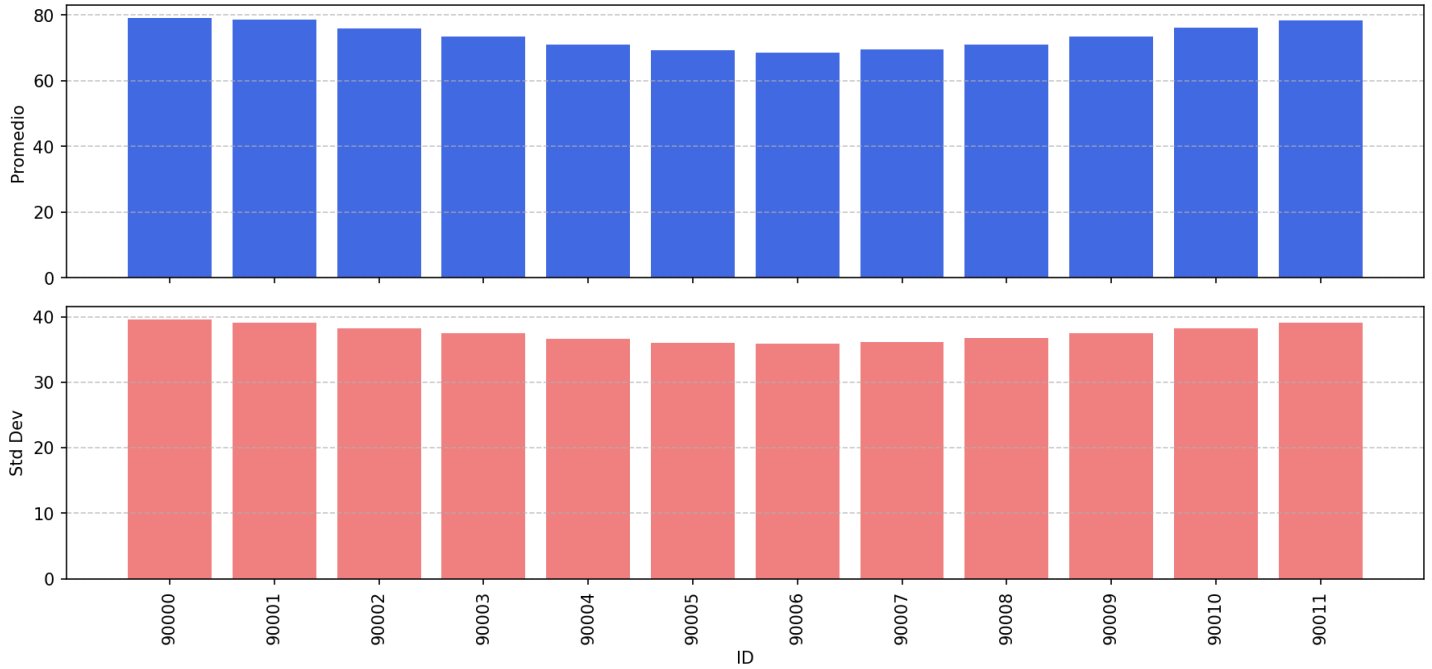
Phi (Azimutal)



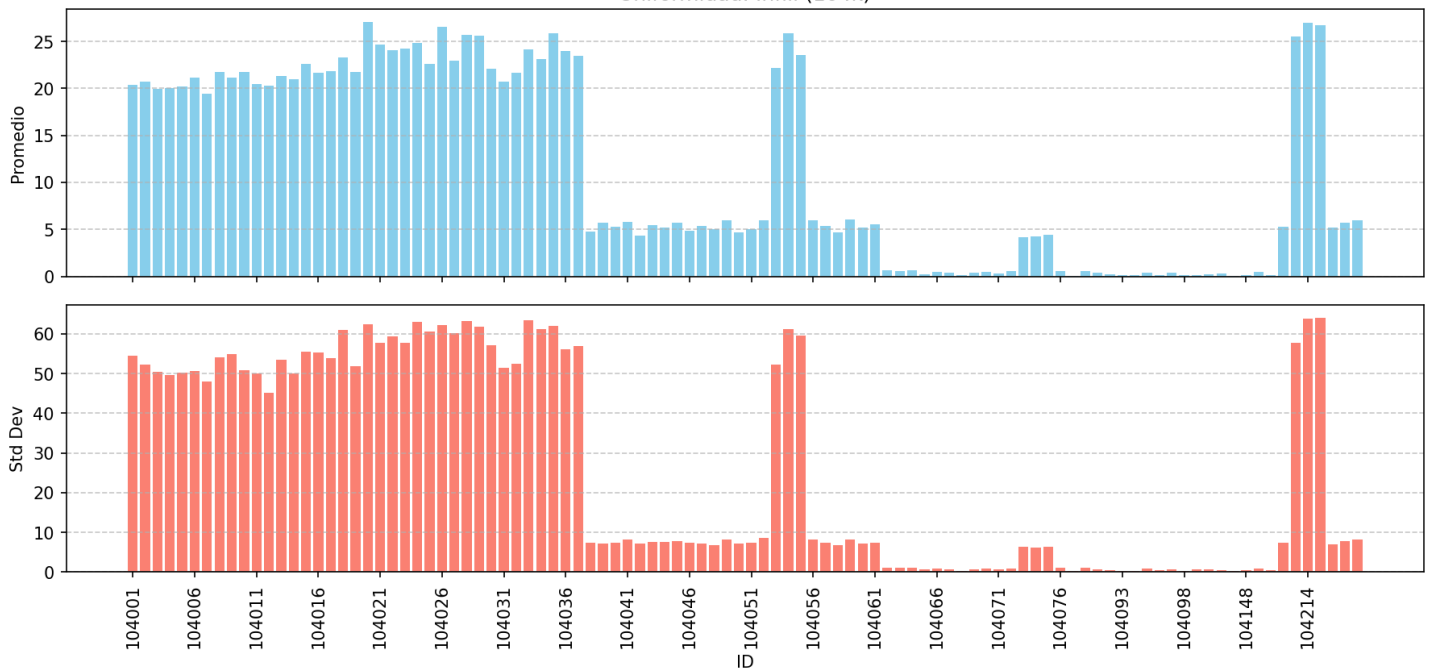
Sesgo (Median): -0.0075 deg
Resolución (MAD): 0.5802 deg
Outliers ($> 15.0^\circ$): 39 (0.17%)

5. Uniformidad por Counter

Uniformidad: Anillo Denso (90k)

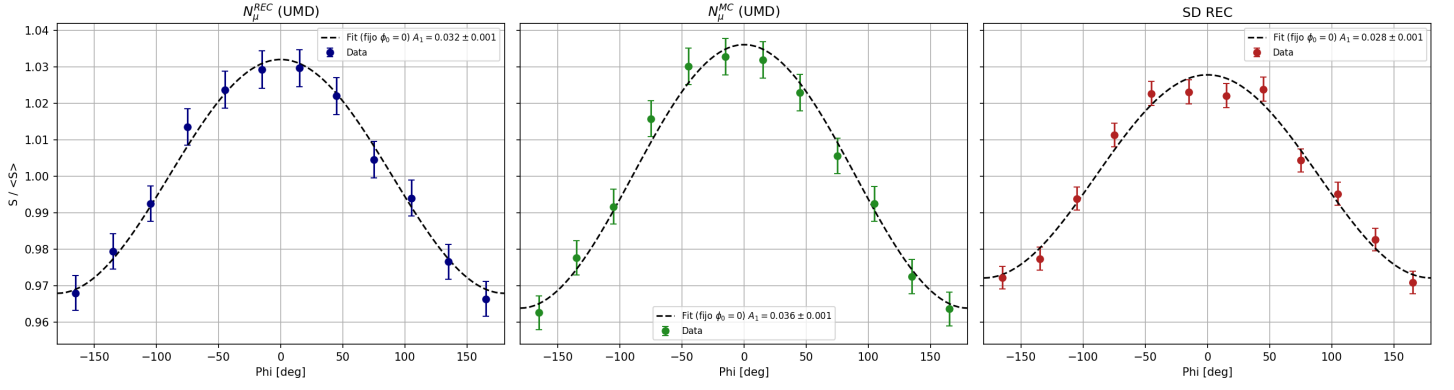


Uniformidad: Infill (104k)

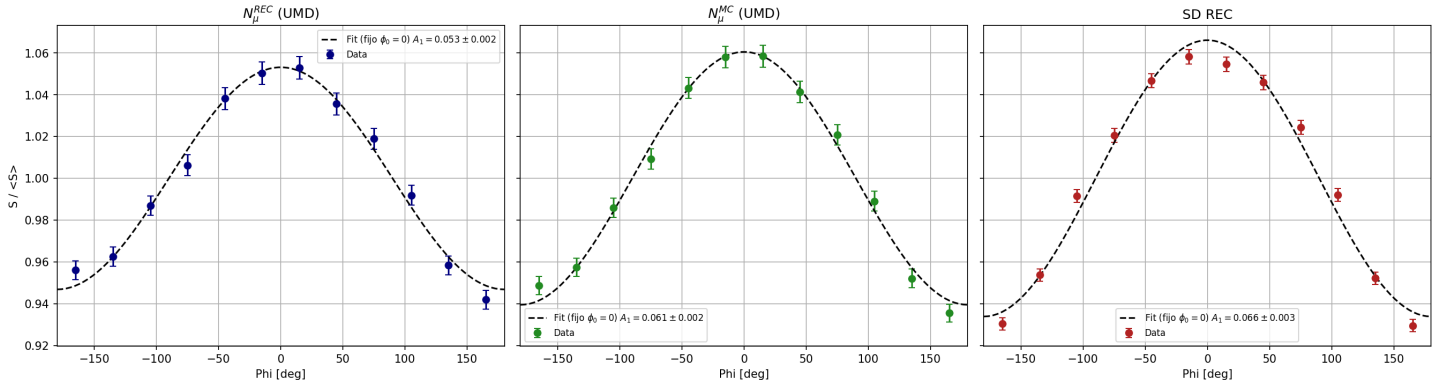


6. Asimetría Azimutal

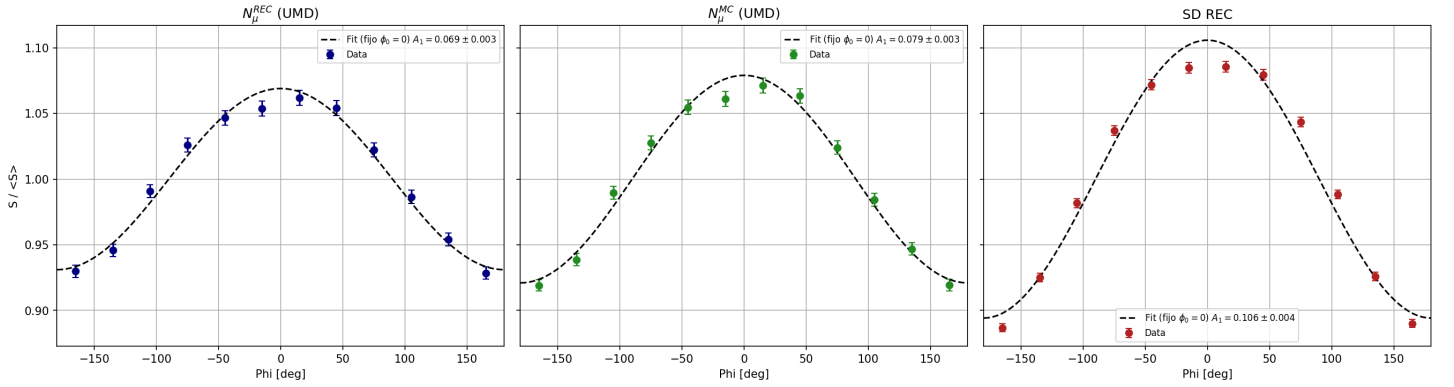
Bin 0: Theta 0.2 - 21.7 deg



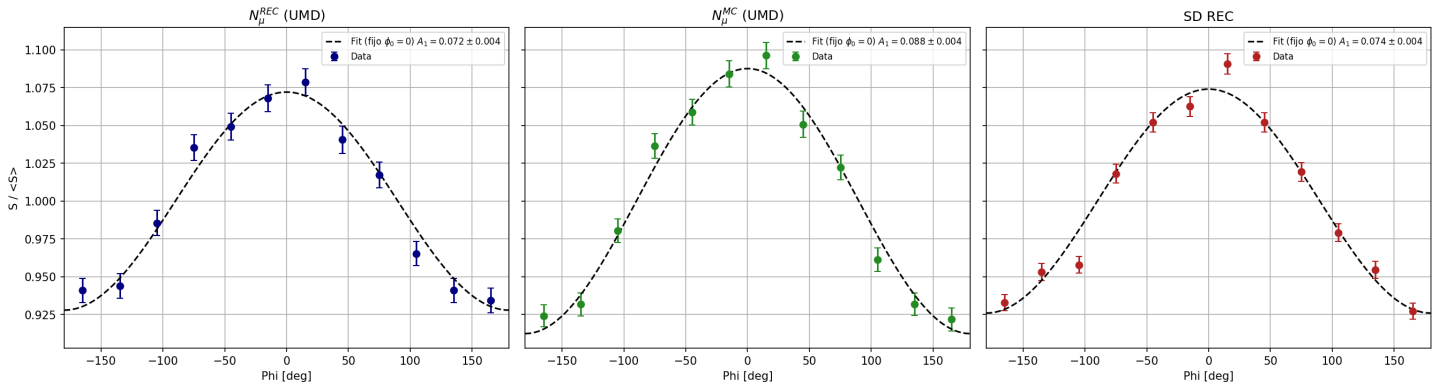
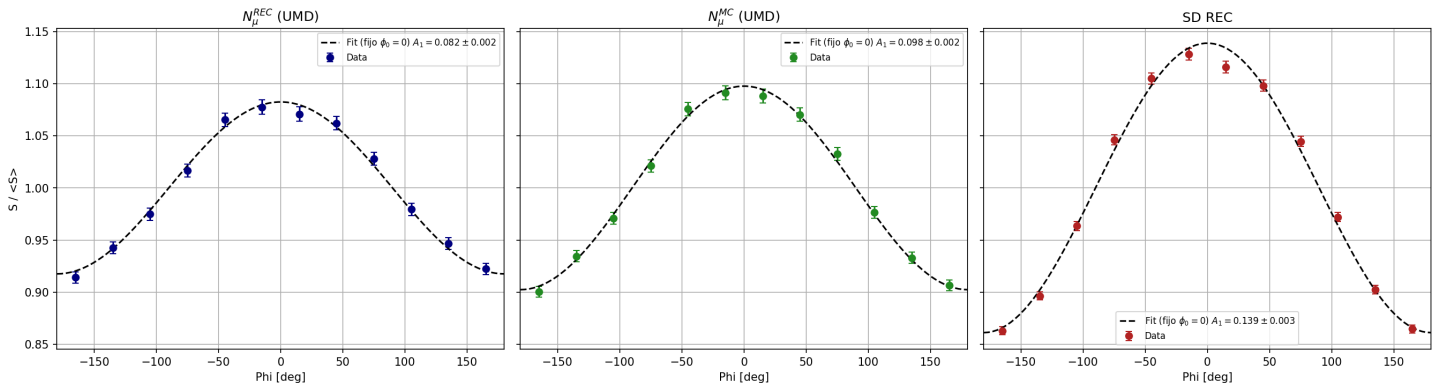
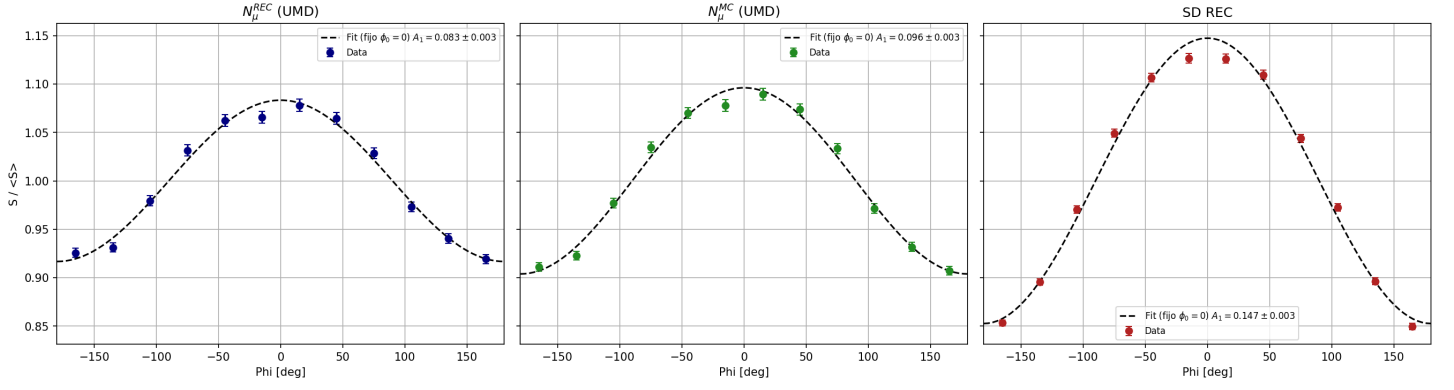
Bin 1: Theta 21.7 - 31.6 deg



Bin 2: Theta 31.6 - 39.9 deg



Bin 3: Theta 39.9 - 47.7 deg



7. Resumen de Ajustes (Comparación)

Valores de Amplitud A1

theta_range	A1_UMD_REC	A1_UMD_MC	A1_SD_REC
0.2-21.7	0.0321	0.0361	0.0279
21.7-31.6	0.0532	0.0605	0.0661
31.6-39.9	0.0690	0.0790	0.1058
39.9-47.7	0.0833	0.0961	0.1474
47.7-55.8	0.0824	0.0976	0.1387
55.8-65.0	0.0721	0.0876	0.0740

